

CURRICULUM VITAE

Rayleigh X. Lei

Center for Statistical Training and Consulting

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RESEARCH INTEREST

My research focuses on developing interpretable Bayesian models with latent structures that more accurately capture phenomena within social science data sets and creating new computational techniques to more efficiently estimate these models. Insights from the geometry of model or data guide this work by informing model specification and motivating new estimation strategies.

EDUCATION

Ph.D. Statistics

University of Michigan

Thesis: Modeling Simplex-valued Data and Latent Structures

Advisor: Professor (Xuan)Long Nguyen

2016-2022

Ann Arbor, MI

B.A. Mathematics (with Honors)

Columbia University

Honors thesis: Generalizing results from Eric Rowland's "A Natural Prime-Generating Recurrence"

2009-2013

New York, NY

POSTDOCTORAL TRAINING

Postdoctoral Fellow; Advisor: Dr. Yajuan Si

Institute for Social Research, University of Michigan

- Improved the validity of Bayesian inference results by enhancing credible interval coverage for small area estimates

8/2024-06/2025

Ann Arbor, MI

Postdoctoral Fellow; Advisor: Dr. Abel Rodriguez

Department of Statistics, University of Washington

- Designed and implemented novel Bayesian ideal point models to improve estimation of ideological preferences from roll call votes and survey data

9/2022-8/2024

Seattle, WA

RESEARCH EXPERIENCES

Statistician (Continuing Specialist - Research)

Center for Statistical Training and Consulting, Michigan State University

- Collaborate with scientists across various research areas to

7/2025-Ongoing

Lansing, MI

tackle key research questions, acquiring new statistical methods as required, developing and implementing models, conducting data analysis, and preparing reports and publications

Graduate Research Assistant; Advisor: Dr. (Xuan)Long Nguyen
Department of Statistics, University of Michigan

9/2016-8/2022
Ann Arbor, MI

- Developed random movement and direction models for changes in simplicial data
- Examined theoretical properties of and developed algorithms to fit general mixtures of probability simplices and tree-based probability simplices
- Applied optimal transport techniques to cluster traffic patterns and analyze these clusters

Graduate Research Assistant; Advisor: Dr. Jun Zhang
Department of Mathematics, University of Michigan

6/2017-7/2017
Ann Arbor, MI

- Investigated connection between information geometry and Hamiltonian and Riemannian Manifold Hamiltonian Monte Carlo

Researcher; Advisor: Dr. Andrew Gelman
Department of Statistics, Columbia University

6/2015-5/2016
New York, NY

- Implemented Bayesian regression models to analyze voter behavior and national pride
- Created unary vectorization testing framework and vectorized unary functions in Stan with template metaprogramming
- Compared inference results from the No U-Turn Sampler (NUTS) to Automatic Differentiation Variational Inference (ADVI)

Undergraduate Researcher; Advisor: Dr. Chris Wiggins
Department of Mathematics, Columbia University

9/2010-5/2011
New York, NY

- Created a Python program to process databases and gather data

Research Intern; Advisor: Dr. Lukas Mueller
Boyce Thompson Institute

1/2011-5/2013 (3 terms)
Ithaca, NY

- Developed web-based tools to facilitate visualization of experimental results using Moose, Catalyst, Mason, and PostgreSQL

OTHER EXPERIENCES

Technical Services, Hospital Billing
Epic Systems

06/2013-05/2015
Verona, WI

- Provided hospital billing software technical support for assigned customers

PUBLICATIONS

1. **Rayleigh Lei** and Abel Rodriguez. 2025. Logistic Unfolding Models for Binary Preference Data. *Statistics and Computing*, 35 (41).
<https://doi.org/10.1007/s11222-025-10570-5>.

2. Sunrit Chakraborty*, **Rayleigh Lei***, and (Xuan)Long Nguyen. 2025. Learning Topic Hierarchies by Tree-Directed Latent Variable Models. *Forthcoming in Bernoulli*. <https://arxiv.org/abs/2408.14327>.
* co-first author
3. **Rayleigh Lei** and Abel Rodriguez. 2025. A Novel Class of Unfolding Models for Binary Preference Data. *Political Analysis*, 33 (1), 32-48. <https://doi.org/10.1017/pan.2024.11>.
4. **Rayleigh Lei** and Abel Rodriguez. 2024. Dynamic Factor Models for Binary Data in Circular Spaces: An Application to the U.S. Supreme Court. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 73 (4), 1042 - 1064. <https://doi.org/10.1093/jrsssc/qlae025>.
5. Sunrit Chakraborty, Aritra Guha, **Rayleigh Lei**, (Xuan)Long Nguyen. 2023. Scalable nonparametric Bayesian learning for dynamic velocity fields. *Proceedings of the Thirty-Ninth Conference on Uncertainty in Artificial Intelligence*, PMLR 216:282-292. <https://proceedings.mlr.press/v216/chakraborty23a.html>.
6. Aritra Guha, **Rayleigh Lei**, Jiacheng Zhu, (Xuan)Long Nguyen, and Ding Zhao. 2022. Robust unsupervised learning of temporal dynamic vehicle-to-vehicle interactions, *Transportation Research Part C: Emerging Technologies*, 142, 103768, <https://doi.org/10.1016/j.trc.2022.103768>.
7. **Rayleigh Lei**, Andrew Gelman, and Yair Ghitza. 2017. The 2008 Election: A Preregistered Replication Analysis. *Statistics and Public Policy*, 4 (1), 1-8. <https://doi.org/10.1080/2330443X.2016.1277966>.

Works under preparations or submitted:

8. **Rayleigh Lei** and Abel Rodriguez. 2026. Modeling Ordinal Survey Data with Unfolding Models. *Submitted to Psychometrika*. <https://arxiv.org/abs/2601.19167>.
9. **Rayleigh Lei** and Yajuan Si. 2026. Calibrating hierarchical Bayesian domain inference for a proportion. *Submitted to Journal of Survey Statistics and Methodology*. <https://arxiv.org/abs/2512.18479>.
10. Marianne Huebner, **Rayleigh Lei**, and Tim Cole. 2026. Comparing weightlifting performance of Masters athletes across age, body mass, and sex from 2000 to 2025. *Revised and resubmitted to International Journal of Sports Physiology and Performance*.
11. Skylar Shi, **Rayleigh Lei**, and Abel Rodriguez. 2026. pumBayes: Bayesian Estimation of Probit Unfolding Models for Binary Preference Data in R. *Under revision for the Journal of Open Research Software*.

arXiv preprints:

12. **Rayleigh Lei** and (Xuan)Long Nguyen. 2024. Modeling Random Direction of Changes in Simplex-valued Data. <https://arxiv.org/abs/2310.19985>.
13. **Rayleigh Lei** and (Xuan)Long Nguyen. 2022. Modeling Random Direction in 2D Simplex Data. <https://arxiv.org/abs/2103.12214>.

Technical reports:

14. **Rayleigh Lei.** Analyzing associations between cross brain region interactions and six aphasia assessments. *Report prepared for Dr. Victoria Diedrichs.* November 2025.
15. Rayleigh Lei and Steven Pierce. COMLEX-USA Level 1 expected pass rate for Class of 2027. *Report prepared for College of Osteopathic Medicine, Michigan State University.* October 2025.
16. Rayleigh Lei and Steven Pierce. COMLEX-USA Level 1 expected pass rate for Class of 2027 Reproducibility log. *Report prepared for College of Osteopathic Medicine, Michigan State University.* October 2025.
17. **Rayleigh Lei.** Two-stage estimator for forestry data. *Report prepared for Dr. Andrew Finley.* August 2025.

PRESENTATIONS

1. **Rayleigh Lei, Yajuan Si.** 2024. FAB Intervals for Binary Outcome Data and Normal Prior. Poster presentation at the 14th International Conference on Bayesian Nonparametrics, Los Angeles, California, June 23-27, 2025.
2. **Rayleigh Lei, Abel Rodriguez.** 2024. Dynamic Factor Models for Binary Data in Circular Spaces: An Application to the U.S. Supreme Court. Invited oral presentation at Joint Statistical Meeting at Portland, Oregon on August 3-8, 2024.
3. **Rayleigh Lei, Abel Rodriguez.** 2024. Dynamic Factor Models for Binary Data in Circular Spaces: An Application to the U.S. Supreme Court. Invited oral presentation at International Society of Bayesian Analysis World Meeting at Venice, Italy on July 1-7, 2024.
4. **Rayleigh Lei, Abel Rodriguez.** 2023. A Novel Class of Unfolding Models for Binary Preference Data. Short oral presentation at BAYSM 2023 on November 13-17, 2023.
5. **Rayleigh Lei, Sunrit Chakraborty, (Xuan)Long Nguyen.** 2023. Geometrically Fitting Tree-directed Topic Models. Poster presentation at BayesComp 2023 on March 15-17, 2023.
6. **Rayleigh Lei and (Xuan)Long Nguyen.** 2021. Modeling Random Directions in 2D Simplex Data. Speed oral presentation at Joint Statistical Meetings on August 6-11, 2021.
7. **Rayleigh Lei and (Xuan)Long Nguyen.** 2021. Modeling Random Directions in 2D Simplex Data. Oral presentation at International Society of Bayesian Analysis World Meeting 2021 on June 28-July 02, 2021.
8. **Rayleigh Lei and (Xuan)Long Nguyen.** 2019. Modeling Simplex Data Transformations (v2). Poster presentation at the Statistics in the Data Science Era: A Symposium to Celebrate 50 Years of Statistics at the University of Michigan, Ann Arbor, MI, September 20-21, 2019.
9. **Rayleigh Lei and (Xuan)Long Nguyen.** 2019. Modeling Simplex Data Transformations (v1). Poster presentation at the 12th International Conference on Bayesian Nonparametrics, Oxford, UK, June 24-28, 2019.

10. Aritra Guha, **Rayleigh Lei**, Jiahui Ji, Jawad Mroueh, and (Xuan)Long Nguyen. 2018. Clustering and Evaluation of Driving Primitives. Poster presentation at the 2018 University of Michigan Toyota Research Institute Annual Review, Ann Arbor, MI, November 13, 2018.

TEACHING EXPERIENCES

Certificates:

University of Michigan Graduate Teacher Certificate 2020-2021

Guest Lecturer, University of Michigan:

- STATS 551: Topics in Bayesian Modeling and Computation 03/2021

Graduate Student Instructor, University of Michigan:

- STATS 408: Statistical Principles for Problem Solving: A Systems Approach Fall 2017
- STATS 250: Introduction to Statistics and Data Analysis Fall 2016
- STATS 412: Introduction to Probability and Statistics Spring 2017
- STATS 499: Honors Seminar Fall 2018, 2019
- STATS 501: Applied Statistics Fall 2021
- STATS 503: Statistical Learning II: Multivariate Analysis Spring 2018
- STATS 511: Statistical Inference Spring 2022

MENTORING EXPERIENCES

Skylar Shi, Masters in Statistics 2024

University of Washington

- Developed R package implementing probit unfolding models

Xinyu He, Junior in Data Science 2020

University of Michigan

- Explored the basics of optimal transportation

Di Wang, Senior in Mathematics, Data Science, and Honors Statistics 2019

University of Michigan

- Utilized mixture model to model how the proportion for a certain income category in Los Angeles County changed yearly from 1990 to 2010

Yingsi Jian, Senior in Mathematics, Data Science, and Honors Statistics 2018-2019

University of Michigan

- Applied topic modeling to analyze chords and voice leading strands in Bach chorales and explore topic modeling with a distance metric on “words”
- Supervised her Senior Honors Thesis (earned the highest honors in statistics)

HONORS AND AWARDS

University of Michigan Postdoctoral Association Travel Award 2025

American Statistical Association Travel Grant 2024

International Society of Bayesian Analysis World Meeting Travel Grant 2024

eScience Postdoctoral Research Fund Recipient 2024

BayesComp Travel Grant	2023
eScience Postdoctoral Research Fund Recipient	2023
University of Washington Data Science Postdoctoral Fellow	2022-2024
Outstanding Teaching Award, <i>Honorable Mention</i> , Department of Statistics University of Michigan	2022
University of Michigan Graduate Teacher Certificate	2021
Junior Travel Support, the 12 th International Conference on Bayesian Nonparametrics	2019
Outstanding Teaching Award, Department of Statistics, University of Michigan	2018
National Science Foundation Graduate Research Fellowship	2018-2023

PROFESSIONAL ACTIVITY

2024 Joint Statistical Meeting, Portland, Oregon

- Organized Invited Program session, “Advances and Applications in Modeling Dynamic Data” 08/2024

University of Washington

- First round reviewer of University of Washington statistics PhD student applications 2022-2023

University of Michigan

- Statistics Graduate Student Justice, Equity, Diversity, and Inclusion committee member 2020-2022
- Co-organizer for Statistics Directed Reading Group 2020-2021
- Statistics PhD Student Council member 2019-2022
- Co-Organizer for 2019 Michigan Student Symposium for Interdisciplinary Statistics Sciences (representing the Department of Statistics) 2018-2019
- Union representative for Department of Statistics’ Graduate Student Instructors 2016-2018